

Multithreading, Concurrency & JVM Internals - Detailed Notes

DETAILED COMPREHENSIVE NOTES

Subject: Java Programming | Chapter: Chapter 3: Advanced Multithreading, Concurrency & JVM Internals | Academic Year: 2026-2027

1. Thread Lifecycle & Thread Creation

A thread is a lightweight unit of execution. In Java, threads transition through states: NEW, RUNNABLE, BLOCKED, WAITING, TIMED_WAITING, and TERMINATED.

Threads can be created by extending the Thread class or implementing the Runnable / Callable interfaces. Implementing Runnable is preferred because Java does not support multiple inheritance, promoting clean separation of task and execution.

2. Synchronization, Locks & Java Concurrency Utilities

The synchronized keyword ensures mutual exclusion using an object monitor lock (intrinsic lock). While effective, synchronized blocks can cause thread starvation and lack timeout capabilities.

The java.util.concurrent package provides ReentrantLock, ReadWriteLock, Semaphore, CountDownLatch, and CyclicBarrier for advanced thread coordination.

Atomic classes (AtomicInteger, AtomicReference) leverage hardware Compare-And-Swap (CAS) instructions to achieve non-blocking lock-free thread safety.

- * volatile keyword: Guarantees visibility across CPU caches by reading directly from main memory and preventing instruction reordering.
- * ExecutorService: Thread pool management abstraction decoupling task submission from thread management (Executors.newFixedThreadPool, newCachedThreadPool).

3. JVM Memory Model & Garbage Collection Algorithms

The Java Memory Model (JMM) defines how threads interact through memory. The Heap is segregated into Young Generation (Eden, Survivor S0, S1) and Old (Tenured) Generation.

Minor GC collects short-lived objects in Young Gen via the Stop-the-World copying collector. Surviving objects are promoted to Old Gen. Major/Full GC reclaims Old Gen memory.

Modern garbage collectors include G1 GC (Garbage-First), ZGC (low-latency concurrent collector), and Shenandoah GC, designed to keep pause times under 10 milliseconds.